

Session 09

CSS Basics – Styling Your Web Pages

What are you going to learn in this session?

In this session, you will begin working with CSS, which is used to control the visual appearance of your web pages. Until now, you have built structured HTML pages, but they likely look plain and unstyled. CSS helps you transform those pages into something visually appealing and user-friendly.

You will learn how to apply styles using different approaches, how selectors work, how to use colors and backgrounds, and how to improve typography. By the end of the session, you will enhance your existing project so it looks more like a real product.

Why is this topic important?

HTML gives structure to your content, but without CSS, your pages will look basic and difficult to use. CSS improves readability, layout clarity, and overall user experience, which are critical for any real-world application.

Separating structure and styling also makes your code cleaner and easier to maintain. This concept, known as separation of concerns, is a fundamental principle in software development that you will continue to use across technologies.

How should you prepare yourself to attend this session?

Before attending this session, ensure that your HTML project from the previous session is ready and working. Open your project in VS Code and keep Live Server running so you can instantly see changes in the browser.

Come prepared to experiment with styles and make visible changes. This session is highly practical, so you should focus on applying styles and observing how they affect the page.

What should you do after the session?

After the session, revisit your project and improve its design further by experimenting with colors, fonts, and layout. Try creating different visual styles and observe how small changes impact readability and user experience.

You should also practice writing selectors and applying styles to different parts of your page until you become comfortable with how CSS targets elements.

Class Notes

Why CSS is Needed

Plain HTML pages are functional but not visually appealing. Without CSS, all text appears similar, spacing is inconsistent, and there is no clear hierarchy, which makes the page difficult to read.

CSS allows you to control how content looks, including colors, spacing, fonts, and layout. It transforms a basic page into something that is easier to read and more engaging for users.

Ways to Add CSS

There are three main ways to add CSS to your project, each serving a different purpose. While all three work, external CSS is the preferred method for real-world projects because it keeps styling separate from structure.

```
HTML
<!-- Inline CSS -->
<p style="color: red;">This is red text</p>

<!-- Internal CSS -->
<style>
  p {
    color: blue;
  }
</style>

<!-- External CSS -->
<link rel="stylesheet" href="styles.css" />
```

In external CSS, you write styles in a separate file like `styles.css` and link it to your HTML. This makes your code cleaner and reusable across multiple pages.

CSS Selectors (Core Skill)

Selectors are used to target HTML elements so that styles can be applied to them.

Understanding selectors is one of the most important skills in CSS because it determines where your styles will apply.

CSS

```
/* Tag selector */
p {
  color: green;
}

/* Class selector */
.highlight {
  color: orange;
}

/* ID selector */
#main-title {
  color: purple;
}
```

You can connect these selectors to your HTML like this:

HTML

```
<h1 id="main-title">Welcome</h1>
<p class="highlight">This is highlighted text</p>
```

Class selectors are reusable, while ID selectors are unique and should be used for a single element.

Grouping and Universal Selectors

Sometimes you want to apply the same style to multiple elements at once. Grouping selectors allow you to do this efficiently, reducing repetition in your code.

CSS

```
h1, h2, p {
  font-family: Arial;
}
```

The universal selector applies styles to all elements on the page and is often used for basic resets or global styling.

```
CSS
* {
  margin: 0;
  padding: 0;
}
```

These patterns help you write cleaner and more maintainable CSS.

Colors in CSS

CSS allows you to apply colors in different formats depending on your preference. The most common approaches include named colors, hexadecimal values, and RGB values.

```
CSS
p {
  color: red;
}

h1 {
  color: #ff0000;
}

body {
  background-color: rgb(240, 240, 240);
}
```

Changing colors is one of the quickest ways to make your page look better and create visual hierarchy.

Background Properties

Background properties allow you to control the background of elements, which helps in separating sections visually and improving design clarity.

```
CSS
body {
  background-color: #f5f5f5;
}

header {
```

```
background-color: #333;  
color: white;  
}
```

You can also add background images, but for now, focus on simple color usage to understand how sections are visually separated.

Typography Basics

Typography controls how text appears on your page, and it plays a major role in readability. Small changes in font size, spacing, and alignment can significantly improve the user experience.

```
CSS  
h1 {  
  font-size: 32px;  
  font-weight: bold;  
  text-align: center;  
}  
  
p {  
  font-size: 16px;  
  font-family: Arial, sans-serif;  
}
```

By applying typography consistently, your page becomes easier to read and more professional.

Applying Styling to Your Project

Now you will apply everything you have learned to your existing project. Focus on improving visible areas such as the header, table, and form so that your page looks structured and visually appealing.

- Add classes to different sections of your HTML
- Apply colors to distinguish sections
- Adjust font sizes and alignment for better readability

The goal is not perfection but noticeable improvement. You should clearly see the difference between your original HTML page and the styled version.

Common Beginner Confusions

Many learners struggle with selectors, especially understanding when to use class versus id. A simple rule is that classes are reusable, while ids are meant for unique elements.

Another common issue is CSS not applying, which is usually caused by incorrect file linking or selector mismatch. Always check whether your CSS file is properly linked and whether the selector matches the HTML element.

Key Takeaways

You now understand how CSS enhances the visual appearance of web pages and improves user experience. You have learned how to apply styles using selectors, define colors, manage backgrounds, and improve typography.

You have also seen how separating HTML and CSS leads to cleaner and more maintainable code. At this stage, focus on experimenting and applying styles rather than memorizing syntax, as practical usage will build your confidence over time.